Milestone-2 (Temporal Trend Analysis of US Disaster Declarations)

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1. Project Description

The **Temporal Trend Analysis of US Disaster Declarations** project focuses on analyzing historical disaster declaration data released by FEMA in the United States. The main goal of the project is to identify patterns in disaster occurrences over time and understand how different disaster types evolve across years and months.

Disasters such as floods, hurricanes, storms, wildfires, and other natural hazards have significant social and economic impacts. By analyzing historical disaster data, it is possible to observe long-term trends, seasonal patterns, and extreme disaster years.

Using Python-based data analysis tools such as Pandas, NumPy, Matplotlib, and Seaborn, the dataset is processed and visualized to extract meaningful insights. The project performs multiple analytical steps including yearly trend analysis, disaster type comparison, seasonal analysis, rolling average trend analysis, growth rate computation, and peak disaster year detection.

This analysis helps transform raw disaster declaration data into understandable patterns that can assist in disaster preparedness and policy planning.

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2. Dataset Used

The dataset used in this project is the **US Disaster Declarations Dataset** published by FEMA (Federal Emergency Management Agency).

The dataset contains historical information about disaster declarations across different states in the United States. Each row represents a disaster declaration event.

Important attributes in the dataset include:

• disaster_number – Unique identifier of the disaster
• state – State where the disaster occurred
• declaration_type – Type of disaster declaration
• declaration_date – Date when the disaster was officially declared
• incident_type – Type of disaster such as flood, hurricane, wildfire, etc.
• incident_begin_date – Start date of the disaster event
• incident_end_date – End date of the disaster event
• designated_area – Specific region affected by the disaster
• region – FEMA administrative region

For temporal analysis, the most important column used is **declaration_date**, which is used to extract time-based features such as year and month.

3. Environment Setup

The project was implemented using **Python in Jupyter Notebook**.

The following libraries were used for data analysis and visualization:

• **Pandas** – Used for data loading, cleaning, and manipulation
• **NumPy** – Used for numerical operations and calculations
• **Matplotlib** – Used for creating graphs and visualizing trends
• **Seaborn** – Used for enhanced data visualization and styling

These libraries allow efficient handling of structured datasets and enable clear visualization of disaster patterns over time.

4. Data Exploration

Before performing the analysis, the dataset was explored to understand its structure and verify that all required fields were available.

The exploration process included:

• Loading the dataset using Pandas
• Displaying the first few rows using `df.head()`
• Inspecting column names using `df.columns`
• Understanding the types of variables present in the dataset

This step ensures that the dataset contains the necessary information required for temporal trend analysis.

5. Data Preprocessing

Data preprocessing is an important step to prepare raw data for analysis. In this project, preprocessing focused mainly on handling the time column and creating new features required for trend analysis.

5.1 Data Loading

The dataset was loaded into a Pandas DataFrame using the read_csv() function. This allows the dataset to be easily manipulated and analyzed using Python.

Once loaded, the first few rows were displayed to confirm that the dataset was imported correctly.

5.2 Date Conversion

The **declaration_date** column originally exists in string format. For performing time-based analysis, it was converted into datetime format using Pandas.

This conversion ensures that Python correctly interprets the date values and allows extraction of time components such as year and month.

Invalid or incorrect date values were handled using the errors='coerce' parameter, which converts invalid entries into missing values.

5.3 Feature Extraction (Year & Month)

After converting the date column into datetime format, two new columns were created:

• **year** – Extracted from the declaration date to analyze long-term disaster trends.

• **month** – Extracted to study seasonal disaster patterns.
These extracted features enable grouping of disaster data by year and month, which is essential for trend and seasonality analysis.

6. Proposed System

The proposed system performs a sequence of data analysis operations to study disaster patterns over time.

The system workflow consists of the following stages:

1. **Dataset Input**
   The FEMA disaster declaration dataset is loaded into the analysis environment.

2. **Data Preprocessing**
   The declaration date is converted to datetime format, and new features such as year and month are extracted.

3. **Temporal Analysis**
   The dataset is grouped and analyzed based on time attributes such as year and month.

4. **Visualization**
   Various graphs and charts are generated to visualize trends and patterns in disaster occurrences.

5. **Insight Generation**
   Key insights such as growth trends, seasonal patterns, and peak disaster years are identified.

This workflow allows transformation of raw disaster data into meaningful analytical insights.

7. Analysis Pipeline

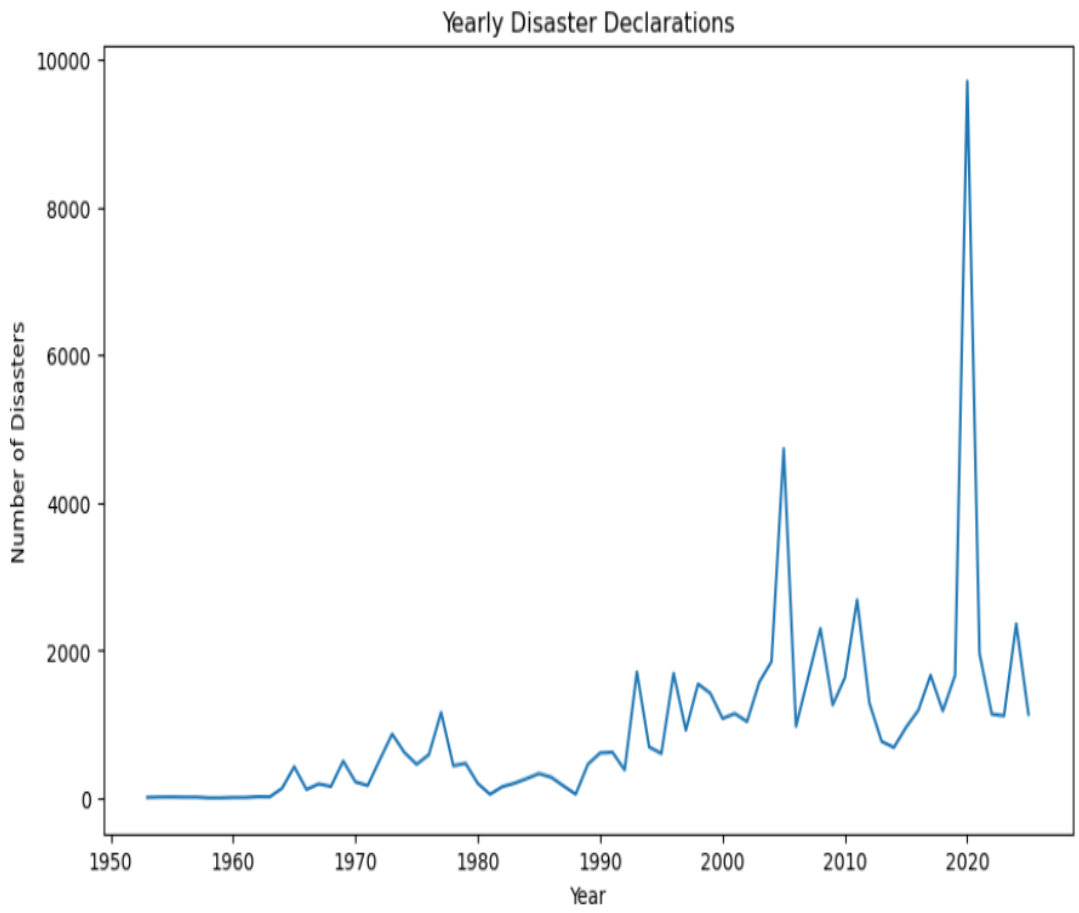
The analysis pipeline includes multiple stages of temporal analysis to understand disaster patterns from different perspectives.

7.1 Yearly Disaster Trend Analysis

To analyze how disaster declarations change over time, the dataset was grouped by year using the Pandas groupby() function.

The total number of disaster declarations for each year was calculated and visualized using a line chart.

This visualization helps identify long-term disaster trends and shows whether disaster declarations are increasing or decreasing over time.



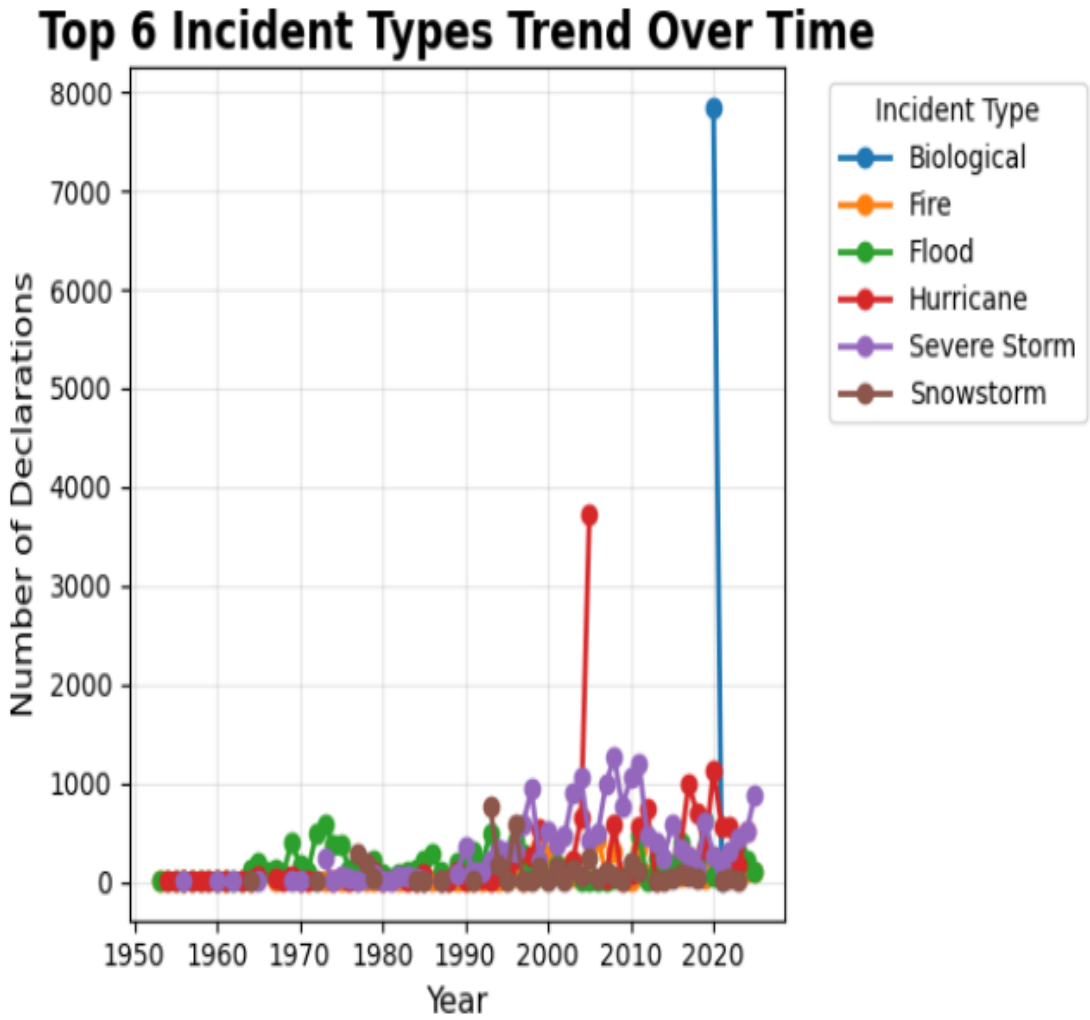
7.2 Incident Type Trend Analysis

Different types of disasters occur in the United States, including floods, hurricanes, storms, and wildfires.

To compare how different disaster types evolve over time:
• The dataset was grouped by **year and incident_type**
• The number of occurrences for each disaster type per year was calculated
• The grouped data was reshaped using unstack() for visualization

A multi-line chart was created where each line represents a different disaster type. This helps identify which disaster types are more frequent and how they change across years.

```
<Figure size 1400x800 with 0 Axes>
```

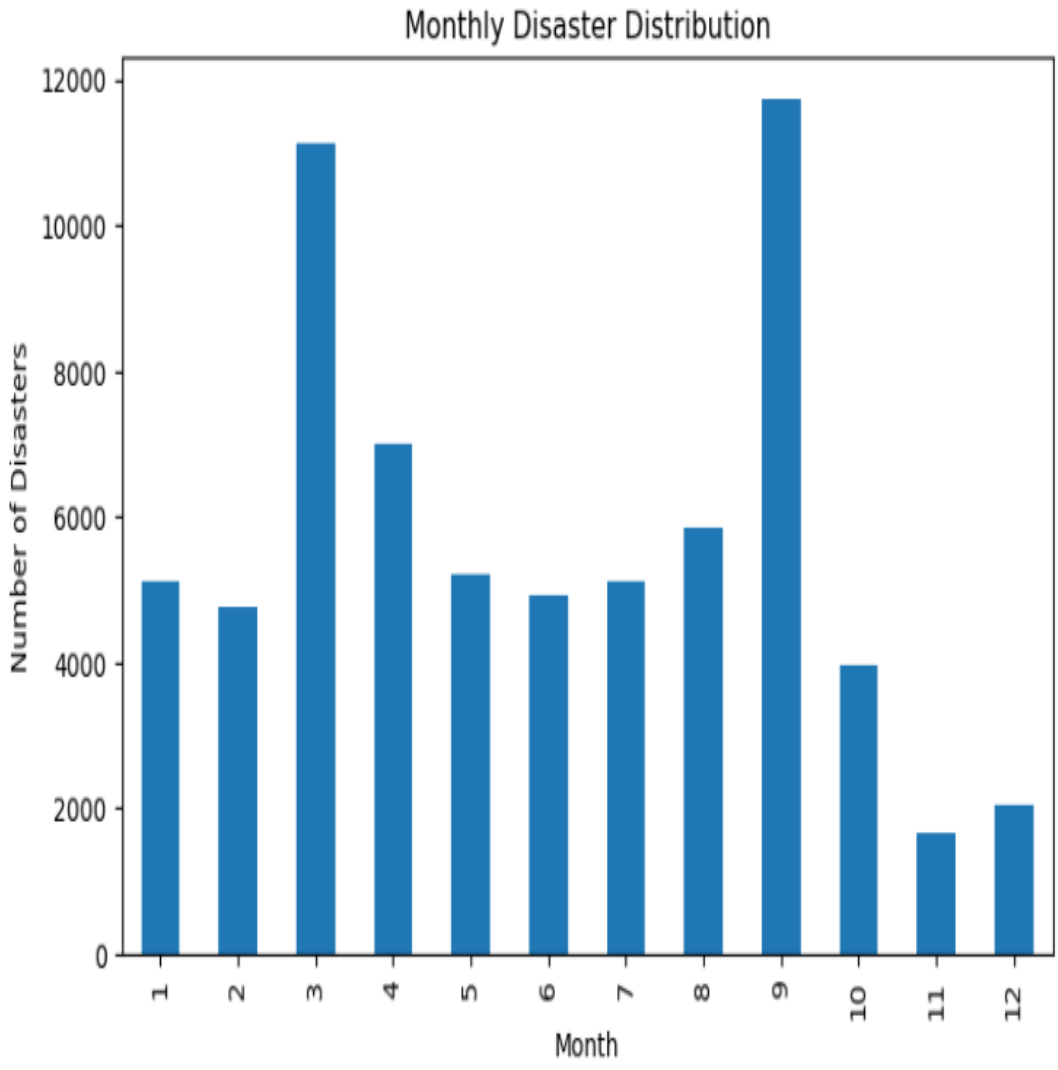


7.3 Monthly Seasonality Analysis

Disasters often occur more frequently during certain months of the year due to seasonal weather patterns.

To analyze seasonality:
• The dataset was grouped by **month**
• The number of disaster declarations per month was calculated
• A bar chart was generated to visualize monthly distribution

This analysis helps identify months with higher disaster activity and seasonal disaster patterns.

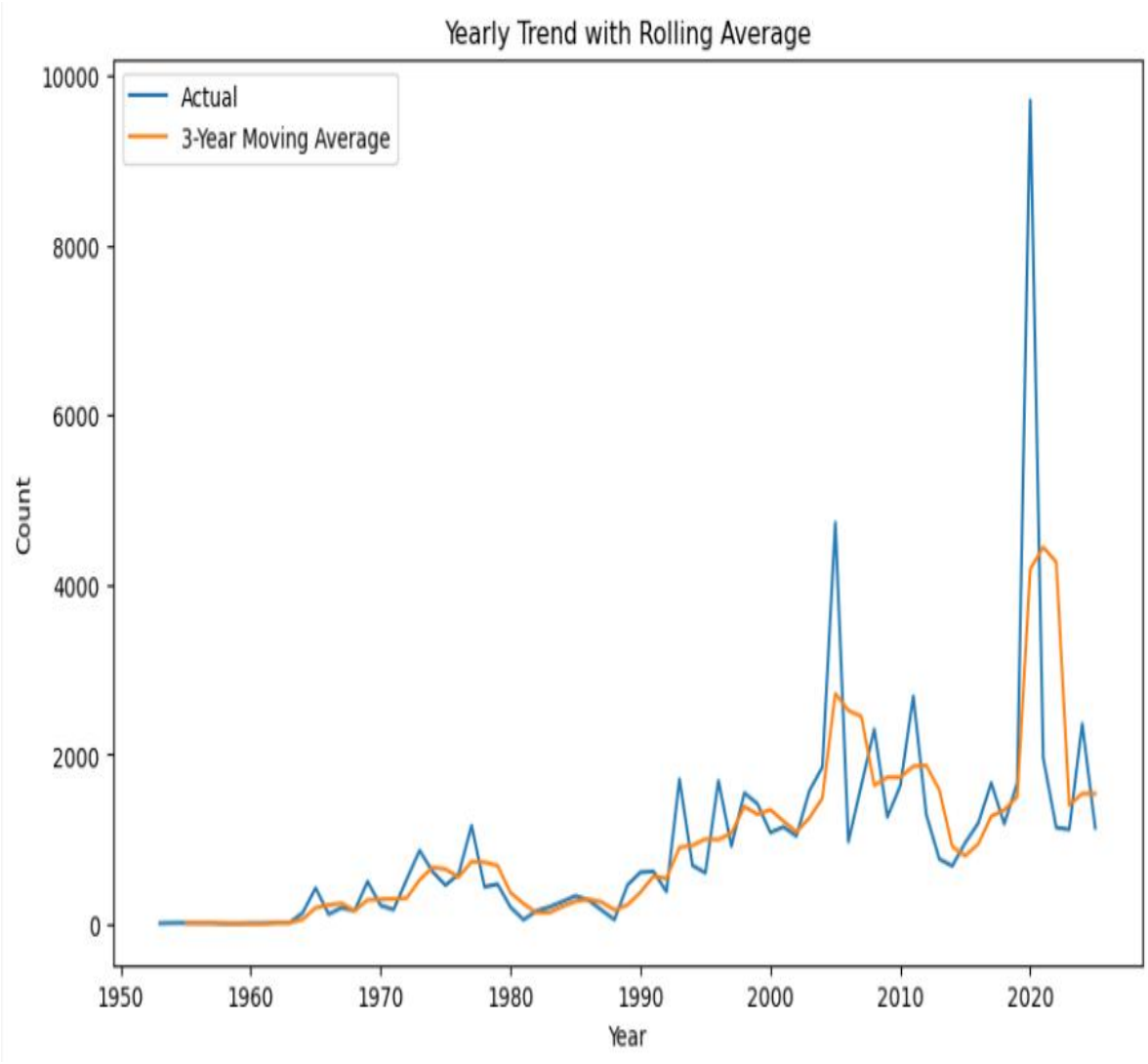


7.4 Rolling Average Trend

Yearly disaster data may contain sudden spikes due to extreme events. To better understand long-term trends, a **3-year rolling average** was calculated.

The rolling average smooths short-term fluctuations and highlights the underlying long-term trend in disaster declarations.

Both the actual yearly trend and the rolling average trend were plotted together to provide better visualization.



7.5 Growth Rate Analysis

The growth rate of disaster declarations was calculated using percentage change between consecutive years.

This shows how disaster frequency increases or decreases from one year to another. Positive values represent an increase in disaster declarations, while negative values represent a decrease.

This analysis helps identify periods of rapid disaster growth.

```
year
1953          NaN
1954    30.769231
1955     5.882353
1956   -11.111111
1957     0.000000
dtype: float64
```

7.6 Peak Year Detection

To identify the years with the highest disaster activity, the yearly disaster counts were sorted in descending order.

The top five years with the highest disaster declarations were selected.

These peak years represent periods with extreme disaster activity and may correspond to major natural disaster events.

```
year
2020    9712
2005    4736
2011    2690
2024    2364
2008    2300
dtype: int64
```

8. Conclusion

The Temporal Trend Analysis of US Disaster Declarations successfully explored historical disaster data to identify patterns across time.

The analysis revealed important insights such as long-term disaster trends, seasonal variations, and differences across disaster types. The study also highlighted years with unusually high disaster activity.

By applying data preprocessing, grouping, and visualization techniques, raw disaster declaration data was transformed into meaningful insights that improve understanding of disaster patterns.

Such analysis can support better disaster management planning and help policymakers prepare for future disaster risks.